

Calculus and Ordinary Differential Equations

AS1109 – Section H

Accelerated Student Lecture Handouts

Lessons 1–4

How to use these notes.

Section H is beginning the course later than the other first-semester sections. These handouts therefore contain the essential definitions, formulas and representative examples already written out.

You are not expected to copy everything from the board.

Use the blank spaces for:

- short explanations;
- diagrams;
- questions;
- additional examples;
- important remarks made during class.

Lesson	Main content
1	Single-variable review; standard graphs; functions of several variables; domain and range
2	Graphs and traces; level curves; contour maps; open/closed and bounded/unbounded regions
3	Limits in two variables; path dependence; estimates; polar coordinates
4	Continuity; first partial derivatives; geometric and physical meaning

The course is accelerated, but the logical order is not being skipped.

The aim is to reduce copying time and combine closely related ideas while keeping the mathematics understandable.

Lesson 1: Functions, standard graphs, several variables, domain and range

Learning goals.

By the end of this lesson you should be able to:

- recall the meanings of function, domain and range;
- recognise the rough graphs of common single-variable functions;
- interpret a function of several variables;
- evaluate $f(x, y)$;
- find domains and simple ranges of functions of two variables.

1. What is a function?

A function assigns exactly one output to every admissible input.

For a single-variable function,

$$y = f(x),$$

the input is a real number x .

The **domain** is the set of admissible inputs.

The **range** is the set of outputs actually produced.

2. Standard single-variable graphs

The following basic graphs should be familiar.

Function	Domain	Range	Main feature
$y = x$	$\mathbb{R}$	$\mathbb{R}$	straight line
$y = x^2$	$\mathbb{R}$	$[0, \infty)$	upward-opening parabola
$y = x^3$	$\mathbb{R}$	$\mathbb{R}$	increasing cubic
$y = \frac{1}{x}$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$	two branches
$y = e^x$	$\mathbb{R}$	$(0, \infty)$	always positive
$y = \ln x$	$(0, \infty)$	$\mathbb{R}$	inverse of e^x
$y = \sin x$	$\mathbb{R}$	$[-1, 1]$	periodic

Three standard domain restrictions are:

$$\sqrt{g(x)} \implies g(x) \geq 0,$$

$$\frac{1}{g(x)} \implies g(x) \neq 0,$$

and

$$\ln g(x) \implies g(x) > 0.$$

Exactly the same restrictions apply when g depends on more than one variable.

3. Functions of several variables

Many quantities in science and engineering depend on several independent variables.

Examples include

$$T = T(x, y),$$

$$P = P(V, T),$$

and

$$\sigma = \sigma(x, y, z).$$

For example, $T(x, y)$ may describe the temperature at the point (x, y) on a thin plate.

A real-valued function of two variables has the form

$$f : D \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}.$$

To every point

$$(x, y) \in D,$$

the function assigns exactly one real number

$$z = f(x, y).$$

Here D is the domain.

Let

$$f(x, y) = x^2 + 2xy - 3y.$$

Then

$$f(2, 1) = 4 + 4 - 3 = 5.$$

Also,

$$f(0, -1) = 3.$$

4. Domains of functions of two variables

The domain is normally taken to be the largest set of real inputs for which the formula gives a real value.

Example 1

$$f(x, y) = \sqrt{9 - x^2 - y^2}.$$

We require

$$9 - x^2 - y^2 \geq 0.$$

Thus

$$x^2 + y^2 \leq 9.$$

The domain is the closed disk of radius 3 centred at the origin.

Example 2

$$g(x, y) = \ln(y - x^2).$$

We require

$$y - x^2 > 0.$$

Therefore

$$y > x^2.$$

The domain is the region lying strictly above the parabola

$$y = x^2.$$

Example 3

$$h(x, y) = \frac{\sqrt{y - x^2}}{x + y - 1}.$$

The square root requires

$$y \geq x^2.$$

The denominator requires

$$x + y \neq 1.$$

Therefore

$$D = \{(x, y) : y \geq x^2, x + y \neq 1\}.$$

When several restrictions occur, **all of them must hold simultaneously**.

The domain is therefore the intersection of the allowed regions.

5. Simple ranges

Consider

$$f(x, y) = x^2 + y^2.$$

Since

$$x^2 + y^2 \geq 0,$$

the function never takes a negative value.

Every non-negative value occurs, so

$$\boxed{\text{Range}(f) = [0, \infty)}.$$

Now consider

$$g(x, y) = 4 - x^2 - y^2.$$

The maximum value is

$$4,$$

attained at

$$(0, 0).$$

There is no lower bound.

Hence

$$\boxed{\text{Range}(g) = (-\infty, 4]}.$$

Exercises for Lesson 1

1. Sketch the rough graphs and state the domains and ranges of

$$y = x^2, \quad y = \frac{1}{x}, \quad y = e^x, \quad y = \ln x.$$

2. For

$$f(x, y) = 3x - 2y + 5,$$

find

$$f(2, -1)$$

and

$$f(0, 0).$$

3. Find and describe geometrically the domain of

$$f(x, y) = \sqrt{16 - x^2 - y^2}.$$

4. Find the domain of

$$f(x, y) = \ln(y - x^2).$$

5. Find the domain of

$$f(x, y) = \frac{1}{\sqrt{9 - x^2 - y^2}}.$$

6. Find the domain of

$$f(x, y) = \frac{x + y}{x - y}.$$

7. Find the domain of

$$f(x, y) = \frac{\sqrt{y - x^2}}{x + y - 1}.$$

8. Find the range of

$$f(x, y) = x^2 + y^2.$$

9. Find the range of

$$f(x, y) = 5 - (x - 2)^2 - (y + 1)^2.$$

10. Find the domain of

$$f(x, y) = \frac{\ln(4 - x^2 - y^2)}{x - 1}.$$

Lesson 2: Graphs, traces, level curves, contour maps and regions

Learning goals.

By the end of this lesson you should be able to:

- interpret

$$z = f(x, y)$$

as a surface;

- find traces of a surface;
- find and interpret level curves;
- read basic information from a contour map;
- classify simple regions as open/closed and bounded/unbounded.

1. Graphs of functions of two variables

The graph of

$$z = f(x, y)$$

is the set of points

$$\{(x, y, z) \in \mathbb{R}^3 : z = f(x, y)\}.$$

Unlike the graph of $y = f(x)$, which is normally a curve in the plane, the graph of $z = f(x, y)$ is normally a **surface in space**.

Examples include:

$$z = x + y \quad (\text{plane}),$$

$$z = x^2 + y^2 \quad (\text{upward-opening paraboloid}),$$

and

$$z = x^2 - y^2 \quad (\text{saddle surface}).$$

2. Traces

A trace is obtained by fixing one independent variable.

If

$$z = f(x, y),$$

then fixing

$$x = a$$

gives

$$z = f(a, y).$$

Fixing

$$y = b$$

gives

$$z = f(x, b).$$

For

$$z = 9 - x^2 - y^2,$$

the trace in the plane

$$x = 0$$

is

$$z = 9 - y^2.$$

The trace in the plane

$$y = 0$$

is

$$z = 9 - x^2.$$

Both are downward-opening parabolas.

3. Level curves

A **level curve** of

$$z = f(x, y)$$

corresponding to the value c is the set of points satisfying

$$f(x, y) = c.$$

Every point on the same level curve has the same function value.

For

$$f(x, y) = x^2 + y^2,$$

the level curves satisfy

$$x^2 + y^2 = c.$$

For

$$c > 0,$$

these are circles of radius

$$\sqrt{c}.$$

For

$$f(x, y) = x^2 - y^2,$$

the level curves satisfy

$$x^2 - y^2 = c.$$

For

$$c = 0,$$

$$x^2 - y^2 = 0$$

gives

$$y = \pm x.$$

For positive and negative values of c , the hyperbolas have different orientations.

4. Contour maps

A contour map displays several level curves at once.

Examples include:

- contour lines of height on a geographical map;
- isotherms on a temperature map;
- pressure contours in fluid mechanics.

Consider

$$T(x, y) = 100 - 5(x^2 + y^2).$$

For

$$T = 80,$$

we obtain

$$x^2 + y^2 = 4.$$

For

$$T = 55,$$

we obtain

$$x^2 + y^2 = 9.$$

For

$$T = 20,$$

we obtain

$$x^2 + y^2 = 16.$$

Thus the corresponding contour radii are

$$2, \quad 3, \quad 4.$$

When consecutive contours represent equal changes in function value:

- closely spaced contours indicate rapid change;
- widely spaced contours indicate slower change.

5. Interior and boundary points

A point of a region is an **interior point** if a sufficiently small disk centred at that point lies completely inside the region.

A **boundary point** is a point for which every sufficiently small disk contains both points belonging to the region and points not belonging to the region.

6. Open, closed, bounded and unbounded regions

A region is **open** if all of its points are interior points.

A region is **closed** if it contains all of its boundary points.

A region is **bounded** if it lies inside some sufficiently large disk.

Otherwise it is **unbounded**.

Examples:

$$x^2 + y^2 < 4$$

is open and bounded.

$$x^2 + y^2 \leq 4$$

is closed and bounded.

$$x > 0$$

is open and unbounded.

Open/closed and bounded/unbounded describe different properties.

For example, a set may be open and bounded.

Exercises for Lesson 2

1. For

$$z = 9 - x^2 - y^2,$$

find the traces in the planes

$$x = 0$$

and

$$y = 0.$$

2. For the same surface, describe the level curve

$$z = c.$$

3. Sketch the level curves

$$c = 0, 1, 4$$

for

$$f(x, y) = x^2 + y^2.$$

4. Describe the level curves

$$c = 0, 1, -1$$

for

$$f(x, y) = x^2 - y^2.$$

5. Describe the level curves of

$$f(x, y) = x + y.$$

6. For

$$T(x, y) = 100 - 5(x^2 + y^2),$$

find the contours

$$T = 80, \quad T = 55, \quad T = 20.$$

Where is the temperature largest?

7. A contour map uses equally spaced values

$$10, 20, 30, 40.$$

The contours are closer together in region A than in region B.

In which region is the function changing more rapidly?

8. Classify

$$\{(x, y) : x^2 + y^2 < 4\}.$$

9. Classify

$$\{(x, y) : 1 \leq x^2 + y^2 \leq 4\}.$$

10. Classify

$$\{(x, y) : 0 < x^2 + y^2 \leq 4\}.$$

Is it open, closed, or neither?

Lesson 3: Limits, path dependence and polar coordinates

Learning goals.

By the end of this lesson you should be able to:

- interpret a limit

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y);$$

- evaluate straightforward limits using algebraic limit laws;
- prove that a limit does not exist using different paths;
- prove simple limits using inequalities;
- use polar coordinates for limits at the origin.

1. Limits in two variables

We write

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$$

if the values of $f(x,y)$ become arbitrarily close to L whenever (x,y) is sufficiently close to (a,b) .

A point in the plane can be approached along infinitely many different paths.
For the limit to exist, all approaches must be consistent with the same limiting value.

2. Direct substitution

Polynomial expressions obey the usual algebraic limit laws.

A quotient may also be evaluated by direct substitution provided the limiting denominator is non-zero.

$$\lim_{(x,y) \rightarrow (1,2)} \frac{x^2 + xy}{x + y} = \frac{1 + 2}{3} = 1.$$

3. The two-path test

If a function approaches two different values along two different paths toward the same point, then the limit does not exist.

Consider

$$f(x, y) = \frac{xy}{x^2 + y^2}.$$

Along

$$y = x,$$

$$f(x, x) = \frac{x^2}{2x^2} = \frac{1}{2}.$$

Along

$$y = -x,$$

$$f(x, -x) = -\frac{1}{2}.$$

Therefore

$$\boxed{\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} \text{ does not exist.}}$$

Checking a few paths and obtaining the same answer does **not** prove that a limit exists.

Different path values prove non-existence.

Existence normally requires a theorem, estimate or squeeze argument.

4. A more subtle path

Consider

$$F(x, y) = \frac{2x^2y}{x^4 + y^2}.$$

Along the family of paths

$$y = kx^2,$$

we obtain

$$F(x, kx^2) = \frac{2k}{1 + k^2}.$$

This depends on k .

Therefore the limit at the origin does not exist.

5. Proving existence using an estimate

Consider

$$G(x, y) = \frac{x^2 y}{x^2 + y^2}.$$

Since

$$0 \leq \frac{x^2}{x^2 + y^2} \leq 1,$$

we have

$$|G(x, y)| = |y| \frac{x^2}{x^2 + y^2} \leq |y|.$$

As

$$(x, y) \rightarrow (0, 0),$$

we have

$$|y| \rightarrow 0.$$

Hence

$$\lim_{(x,y) \rightarrow (0,0)} G(x, y) = 0.$$

6. Changing to polar coordinates

Near the origin, write

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Then

$$r^2 = x^2 + y^2.$$

Approaching

$$(0, 0)$$

means

$$r \rightarrow 0.$$

Consider

$$H(x, y) = \frac{x^2 y^2}{x^2 + y^2}.$$

Substituting

$$x = r \cos \theta, \quad y = r \sin \theta,$$

gives

$$H = r^2 \cos^2 \theta \sin^2 \theta.$$

Since

$$0 \leq \cos^2 \theta \sin^2 \theta \leq 1,$$

we have

$$|H| \leq r^2.$$

Therefore

$$\lim_{(x,y) \rightarrow (0,0)} H(x, y) = 0.$$

A useful pattern is:

if

$$|f(r, \theta)| \leq Cr^\alpha$$

for some

$$\alpha > 0$$

independently of θ , then

$$f(r, \theta) \rightarrow 0$$

as

$$r \rightarrow 0.$$

Exercises for Lesson 3

1. Evaluate

$$\lim_{(x,y) \rightarrow (1,2)} (x^2 + xy + y^2).$$

2. Evaluate

$$\lim_{(x,y) \rightarrow (1,-1)} \frac{x+y}{x^2 + y^2 + 1}.$$

3. Determine whether

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$$

exists.

4. Determine whether

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

exists.

5. Determine whether

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y}{x^4 + y^2}$$

exists.

6. Evaluate

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^2 + y^2}.$$

7. Evaluate

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^2}.$$

8. Use polar coordinates to evaluate

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y^2}{x^2 + y^2}.$$

9. Evaluate

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}}.$$

10. Determine whether

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$$

exists.

Lesson 4: Continuity and first partial derivatives

Learning goals.

By the end of this lesson you should be able to:

- test continuity at a point;
- recognise common continuous functions;
- define first partial derivatives;
- calculate f_x and f_y ;
- interpret partial derivatives as slopes and rates of change.

1. Continuity

A function $f(x, y)$ is **continuous at** (a, b) if:

(i) $f(a, b)$ is defined;

(ii)

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y)$$

exists;

(iii)

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

Polynomials are continuous everywhere.

Rational functions are continuous wherever their denominators are non-zero.

Sums, products and compositions of continuous functions are continuous wherever the resulting formulas are defined.

2. A piecewise example

Consider

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

From Lesson 3,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0.$$

Also,

$$f(0, 0) = 0.$$

Therefore

$$f \text{ is continuous at } (0, 0).$$

3. Partial derivatives

When one variable changes while the other is held fixed, we obtain a **partial derivative**.

For

$$z = f(x, y),$$

the partial derivative with respect to x at (a, b) is

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}.$$

Similarly,

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}.$$

Other common notations are

$$f_x, \quad \frac{\partial f}{\partial x},$$

and

$$f_y, \quad \frac{\partial f}{\partial y}.$$

4. Calculating partial derivatives

Let

$$f(x, y) = x^2y + 3xy^2.$$

When differentiating with respect to x , treat y as a constant:

$$f_x = 2xy + 3y^2.$$

When differentiating with respect to y , treat x as a constant:

$$f_y = x^2 + 6xy.$$

5. Geometric meaning

For the surface

$$z = f(x, y),$$

hold

$$y = b$$

fixed.

The resulting curve is

$$z = f(x, b).$$

The number

$$f_x(a, b)$$

is the slope of the tangent line to this trace at

$$x = a.$$

Similarly,

$$f_y(a, b)$$

is the slope of the trace obtained by holding

$$x = a$$

fixed.

6. Physical interpretation

Suppose

$$T(x, y)$$

is temperature.

Then

$$T_x$$

measures the local rate of temperature change in the x -direction, while

$$T_y$$

measures the corresponding rate in the y -direction.

Let

$$T(x, y) = 100 - x^2 - 2y^2.$$

Then

$$T_x = -2x, \quad T_y = -4y.$$

At

$$(1, 2),$$

$$T_x(1, 2) = -2,$$

and

$$T_y(1, 2) = -8.$$

Both are negative, so the temperature decreases in both positive coordinate directions.

The larger magnitude of T_y shows that the decrease is more rapid in the y -direction at this point.

7. Looking ahead

The coordinate directions are not the only directions in which one can move on a surface.

Later we shall study rates of change in an arbitrary direction.

This leads to:

directional derivatives

and

the gradient.

We shall also differentiate partial derivatives again to obtain

$$f_{xx}, \quad f_{xy}, \quad f_{yx}, \quad f_{yy}.$$

Exercises for Lesson 4

1. Explain why

$$f(x, y) = x^3 - 2xy + y^4$$

is continuous everywhere.

2. State where

$$f(x, y) = \frac{x + y}{x - y}$$

is continuous.

3. State the domain on which

$$g(x, y) = \sqrt{9 - x^2 - y^2}$$

is continuous.

4. Determine whether

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous at the origin.

5. Determine whether

$$g(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous at the origin.

6. Find f_x and f_y for

$$f(x, y) = x^2 y + 3xy^2.$$

7. Find f_x and f_y for

$$f(x, y) = e^{xy} + \sin(x + y).$$

8. Find f_x and f_y for

$$f(x, y) = \ln(x^2 + y^2),$$

and state where these formulas are valid.

9. Find f_x and f_y for

$$f(x, y) = \sqrt{25 - x^2 - y^2}$$

at points where

$$x^2 + y^2 < 25.$$

10. For

$$T(x, y) = 100 - x^2 - 2y^2,$$

find

$$T_x(1, 2)$$

and

$$T_y(1, 2),$$

and explain what their signs and magnitudes mean.

Checklist after Lessons 1–4

You should now be able to:

- recognise the main standard single-variable graphs;
- find domains and simple ranges of functions of two variables;
- interpret the graph

$$z = f(x, y)$$

as a surface;

- distinguish between a trace and a level curve;
- read basic qualitative information from contour maps;
- classify simple regions as open/closed and bounded/unbounded;
- explain why two-variable limits involve many possible paths;
- use two different paths to prove that a limit does not exist;
- use inequalities or polar coordinates to prove some limits exist;
- test continuity at a point;
- calculate and interpret

$$f_x \quad \text{and} \quad f_y.$$

Bring these handouts to lectures and tutorials. Use the blank spaces for short annotations rather than copying the board word-for-word.